Rohith Bayya

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Data Engineer | Data Analyst | Analytics Engineer | AI/ML

SUMMARY

Data professional with 3+ years across data engineering, analytics and applied AI, and an M.S. in Data Analytics Engineering from George Mason University. Builds production pipelines in Python and SQL, governed KPI frameworks and executive dashboards, and statistical models from forecasting and segmentation to controlled experiments. On the AI side, builds RAG and agentic LLM systems with retrieval, guardrails and evaluation harnesses, and ships streaming data platforms on AWS with Spark, Kafka, Docker, Terraform and CI/CD.

TECHNICAL SKILLS

Languages & Querying: SQL, Python, window functions, CTEs, stored procedures, query optimization, performance tuning
Data Engineering: ETL/ELT, data pipelines, dbt, Apache Airflow, Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, streaming, Delta Lake, medallion architecture, dimensional modeling, REST APIs, FastAPI
Warehouses & Cloud: Snowflake, PostgreSQL, DuckDB, SQLite, AWS (S3, EC2), Docker, Terraform, CI/CD, GitHub Actions, Git
AI & Machine Learning: scikit-learn, logistic regression, classification, model validation, feature engineering, LLMs, RAG, FAISS, LangChain, embeddings, semantic search, Text2SQL, function calling, multi-agent orchestration, evaluation harnesses, hallucination guardrails, Ollama, Llama 3.2, AI-assisted development (Claude)
Analytics & Statistics: A/B testing, experimentation, statistical validation, ARIMA forecasting, time series, K-means, segmentation, anomaly detection
BI & Visualization: Power BI, Tableau, Streamlit, Advanced Excel, dashboarding, self-serve analytics, reporting automation, workflow automation
Data Quality & Governance: data quality, schema validation, data governance, KPI governance, data lineage, data documentation, observability, alerting, data cleaning
Practices: stakeholder reporting, executive communication, root cause analysis

EXPERIENCE

Data & AI Engineer (Co-op), SciEncephalon, Remote — Jan 2025 - Dec 2025

- Designed and deployed a Python ingestion and transformation pipeline processing 10,000+ documents per cycle with schema normalization and business rule validation, cutting manual reporting time 60% and lifting pipeline throughput 40%.
- Developed retrieval components for internal document intelligence tooling using recursive chunking, embedding generation, metadata filtering and top-k retrieval, enabling natural language search over enterprise documents.
- Implemented per stage failure logging, anomaly monitoring checks and iterated validation rules across all production cycles, reaching zero undetected pipeline failures during the monitored period.
- Built a Streamlit self serve analytics interface that let non-technical stakeholders resolve their own data requests, eliminating a recurring ad hoc request backlog without added engineering overhead.

Data Analyst, XIT Solutions, Hyderabad, India — Jan 2022 - Dec 2023

- Served as single source of truth for commercial KPIs across 3 business units (Finance, Risk, Compliance), establishing a KPI governance framework with validation stored procedures, metadata documentation and field level lineage behind 50+ downstream reports used by 15 senior leaders.
- Built and maintained 15+ Power BI executive dashboards covering daily operational, monthly (MBR) and quarterly (QBR) review cadences for Finance, Risk and Compliance leadership.
- Authored 12+ SQL query optimizations using window functions, CTEs and indexed joins over 10M+ record multi source pipelines, cutting load time 45%, and saw them adopted as team coding standards.
- Automated 15+ recurring reporting workflows in SQL and Python, compressing turnaround from 4 hours to 30 minutes, an 87% reduction, and driving manual errors to near zero.
- Designed and executed a Frequentist A/B experiment across 8 product categories with a 6 month holdout, achieving a 25% stockout reduction that held at 18-20% after the test, and translated the findings into commercial recommendations.
- Built ARIMA forecasting models for quarterly revenue, margin and cost at 8-12% MAPE on a rolling 3 month holdout, supporting a $2M+ annual budget planning cycle across 3 business units.
- Applied K-means segmentation across 200+ client accounts, producing 3-5 clusters whose resulting profiles the commercial team adopted for its targeted outreach strategy.
- Investigated an intercompany transfer pricing discrepancy that had propagated undetected across 3 reporting systems for 2 quarters, synthesized the findings into an executive summary for leadership, and authored a remediation runbook.

PROJECTS

CrisisLens | Real-Time Disaster Data Platform / Geo-Aware Agentic RAG

- Streamed live USGS earthquake and NOAA/NWS weather feeds into Kafka with an idempotent producer (acks=all), normalizing two heterogeneous sources into one canonical schema with restart safe deduplication and zero duplicate events verified across restarts.
- Engineered a Spark Structured Streaming medallion lakehouse (bronze, silver, gold) on Delta Lake with exactly once checkpointing and latest version deduplication through Delta MERGE upserts.
- Built a geo aware agentic RAG service with spatial pre filtering, FAISS semantic retrieval over local MiniLM embeddings and grounded, cited answers from a local Llama 3.2, with guardrails constraining answers to retrieved context and an evaluation approach measuring groundedness and hallucination rate.
- Deployed a live end to end platform on AWS, provisioning EC2 with Terraform and shipping a Docker image through GitHub Actions CI/CD to GHCR, serving a public hazard API and Leaflet map.

Job Radar | Automated Multi-Source Data Pipeline

- Built a production ETL pipeline ingesting from 15+ job board APIs and thousands of ATS boards hourly through GitHub Actions CI/CD, with deterministic deduplication, incremental processing and multi stage data quality filtering.

Cloud-Native Analytics Lakehouse | dbt, Snowflake, Airflow

- Built an end to end analytics stack from AWS S3 through PySpark to a Snowflake medallion architecture, transformed with dbt (incremental models, schema tests, CI/CD) and scheduled with Airflow DAGs over 1M+ records, with dimensional fact and dimension modeling throughout and a 60% cut in query latency.

ClimateGPT Fusion | Multi-Agent LLM Analytics

- Designed an agentic router dispatching questions to RAG, Text2SQL over DuckDB or direct LLM answering through function calling schemas with SQL validation before execution, and built an evaluation harness comparing correctness, groundedness and hallucination risk across model versions, for 3x faster research turnaround.

Internal Knowledge Assistant | RAG Pipeline

- Built an end to end RAG pipeline over a public document corpus into FAISS indexed chunks, compared OpenAI and Hugging Face sentence transformer embeddings on a held out query set, and tuned chunk size, overlap and retrieval parameters to deliver source grounded Q&A with inline citations.

Carvana Risk Analytics | ML Classification Pipeline

- Built an end to end logistic regression pipeline in Python for risk classification, with feature importance analysis, model validation and performance metrics.

CAIIEC AI Workforce Capstone | George Mason / Schar School

- Cleaned 60K raw records from REST APIs (Adzuna, Jooble, USAJobs) down to 11K validated records using Python and SQL, an 80% noise reduction, and built a Streamlit self serve dashboard for 10+ government researchers, with findings published in a CAIIEC national AI workforce policy brief.

EIA Grid Monitor | Automated Operational KPI Reporting

- Built an Airflow scheduled Python pipeline into PostgreSQL feeding hourly Tableau KPI dashboards, with threshold based alerting and a deviation index.

MarketPulse | Revenue & Channel Analytics

- Built a SQL and Python pipeline from REST APIs with time series analysis and channel level performance decomposition, surfacing revenue pacing and competitive benchmarking through Tableau dashboards.

NYC Crime Analytics | Tableau Executive Dashboards

- Analyzed multi year public datasets to surface trends and delivered Tableau dashboards with five data driven recommendations, focused on executive storytelling and visualization best practice.

EDUCATION

George Mason University, Fairfax, VA - M.S. Data Analytics Engineering

Neil Gogte Institute of Technology, Hyderabad, India - B.Tech Computer Science